

SECTION A: Answer ALL the questions (80 marks) (CLO 1, CLO 2)

1. List FOUR (4) examples of secondary storage. (2 Marks)

Hand disks
Floppy disks
CD ROMs
USB flash drives

2. List THREE (3) levels of programming language. (3 Marks)

Machine
Assembly
High

3. Create a flowchart to simulate a traffic light system. User will enter an input either 'R', 'Y' or 'G' and the system will display 'Red', 'Yellow', and 'Green' respectively. (9 Marks)

4. Give TWO (2) examples for each of the following data types in C. (3 Marks)

a) Primitive data type

→ int, float

b) Derived Types

→ array, pointer, function

c) User Defined Data Types

→ struct, enum, union

5. Write the output for the following C code. (4 Marks)

```
#include <stdio.h>
int main() {
    int a = 5;
    int b = 7.9;
    double c = 7.3;
    char d = 'T';
    printf("The value of A is %5d", a);
    printf("\nThe value of B is %d", (a+b));
    printf("\nThe value of C is %5.3f", c);
    printf("\nThe value of D is %c", d);
    return 0;
}
```

→ space

→ 3 decimal point

6. Determine the **VALID** and **INVALID** identifiers in the following table. Please ensure that all answers are written in the provided answer booklet. (8 Marks)

No	Identifiers	VALID or INVALID
1	Do	INVALID
2	break	VALID
3	Void	INVALID
4	ch@r	INVALID
5	Else	INVALID
6	Double	INVALID
7	dou_ble	INVALID
8	2Numbers	INVALID

7. Given the following C code.

```

1. #include <stdio.h>
2. int main()
3. {
4.     int var = 3;
5.     if (var == 1)
6.         printf("Case 1 is Matched.");
7.     else if (var == 2)
8.         printf("Case 2 is Matched.");
9.     else if (var == 3)
10.        printf("Case 3 is Matched.");
11.    else
12.        printf("Default case is Matched.");
13.    return 0;
14. }
```

- a) Convert the **if...else** statement in the above code to **switch** statement. (6 Marks)

- b) What is the output of the above code if the statement in line 11 is removed? (2 Marks)

for (int i = 0; i <= 20; i++)
if (i % 2 != 0)
 printf("%d", i);

8. Write a **while** statement to display an odd number between 0 to 20.

(5 Marks)

9. Write a **nested for** statement to display the following output.

(5 Marks)

```
1x1=1
1x2=2
1x3=3
1x4=4
1x5=5
1x6=6
```

```
2x1=2
2x2=4
2x3=6
2x4=8
2x5=10
2x6=12
```

Outer loop (2)

Inner loop (6)

10. Complete the following C code to find a squareroot of a number. Give the function name as findSquareRoot. Please ensure that all the answers are written in the provided answer booklet.

(5 Marks)

```
#include <stdio.h>
```

```
#include _____ (1) math.h
```

```
// Create a function with one integer parameter
```

```
void findSquareRoot (int num) (2) _____
```

```
// call built in squareroot function
```

```
int y= sqrt (num) (3) _____;
```

```
printf("The squareroot o/d ' (4) y _____ );
```

```
}
```

```
int main() {
```

```
// call the user-define function
```

```
_____ (5) findSquareRoot (81)
```

```
return 0;
```

```
}
```

Note : The output is "The square root of 81 is 9"

11. Write a user-defined function based on the given information.

Function name : readData
Argument : no argument
Code in body : read one integer value
Return a value : no

(3 Marks)

12. Complete the following C code to display the output as below. Please ensure that all answers are written in the provided answer booklet.

```
arr1 in reverse order
50 40 30 20 10

Multiplication of arr1 and arr2
10 40 90 160 250
```

(6 marks)

```
int main()
{
    int arr1[5] = {10, 20, 30, 40, 50};
    int arr2[5] = {1, 2, 3, 4, 5};
    int arr3[5];
```

```
    printf("arr1 in reverse order \n");
    // display arr1 in reverse order using for statement
```

```
    for(int i = 4; i >= 0; i--) {
```

```
        printf("%d", arr1[i]);
```

```
    }
```

```
    printf("\n");
```

```
    // multiply arr1 and arr2 and store it into arr3
    for (int i = 0; i < 5; i++) {
```

```
        arr3[i] = arr1[i] * arr2[i];
```

```
    printf("\nMultiplication of arr1 and arr2 \n");
```

```
    // display arr3 using for statement
```

```
    for (int i = 0; i < 5; i++) {
        printf("%d", arr3[i]);
    }
```

```
    return 0;
}
```


13. What is the output based on the following C code.

```
#include <stdio.h>
int main()
{
    char arr1[] = {'B', 'u', 'k', 'h', 'a', 'r', 'y'};
    int arr2[] = {'1', '2', '3', '4', '5', '6', '7'};
    double arr3[] = {23.5, 34.6, 45.679, 862.234};

    printf("sizeof arr1 is :%lu ", sizeof(arr1));
    printf("\nsizeof arr2 is :%lu ", sizeof(arr2));
    printf("\nsizeof arr3 is :%lu ", sizeof(arr3));

    int numOfElement = sizeof(arr3)/sizeof(arr3[1]);
    printf("\nNumber of elements is:%d ", numOfElement);
    return 0;
}
```

(4 Marks)

14. Answer the question based on the following information.

Structure name : Student
Structure members : Name : String
Matric No : String
Mark : Integer

a) Declare a structure in C language based on the given information.

(2 Marks)

b) Create a structure variable in C language.

(2 Marks)

c) Assign the following values to the struct variables.

Hint: use strcpy() to assign a string value to any variable

Name : Baharudin
Matric No : 23214
Mark : 87

d) Write a code to display name, matric no and mark.

(3 Marks)

(3 Marks)

15. Given the following C code.

```
int main ()
{
    int i=3;
    int a[5] = {1,2,3,4,5};
    int *p = a;
    return 0;
}
```

a. What is the output when the below statement is executed?

(2 marks)

```
printf("%d", *p+2);
```

$1+2=3$

b. What is the output when the following statement is executed?

(1 marks)

```
printf("%d", a[i]);
```

$=4$

c. What is the output when this statement is executed?

(2 marks)

```
printf("%d", *(a+1));
```

$=2$

20 marks ✓

SECTION B: Answer all question given (20 marks) (CLO 1, CLO 2)

1. Answer the question based on the following information.

You are tasked by your supervisor to solve a problem involving reading temperature values in Celsius from a file named **fdata.txt**. Your task is to convert each value to Fahrenheit and display the output as demonstrated below. Use the following formula to perform the conversion from Celsius to Fahrenheit.

$$\text{Fahrenheit} = 1.8 * \text{Celsius} + 32.0$$

File input : **fdata.txt**

fdata.txt X
56
45
34
56
45
67
86
34
65
86

The output:

Celcius	Fahrenheit
56	132.80
45	113.00
34	93.20
56	132.80
45	113.00
67	152.60
86	186.80
34	93.20
65	149.00
86	186.80

- a. Draw the flowchart based on the above scenario.

(7 marks)